

Hexadecimal & Colour Codes

A Guide to Understanding Colors in the Digital World

1. The Digital World of Pixels

Everything you see on computer screens is made of tiny dots called **pixels** [cite: 3, 5]. Each pixel glows with digital colors created from light [cite: 5]. Understanding these colors is the first step to creating amazing things with computers and AI [cite: 6].

2. What are Hexadecimal Numbers?

Hexadecimal is a special way of counting using **16 digits** instead of the usual 10 [cite: 10].

- **Regular Digits:** 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 [cite: 11]
- **Hexadecimal Additions:** A, B, C, D, E, F [cite: 12]

Think of it as counting beyond 9! When we run out of single digits, we use letters [cite: 31, 32]:

A = 10, B = 11, C = 12, D = 13, E = 14, F = 15 [cite: 33]

3. Anatomy of a Hex Color Code

The **#** symbol acts as a special signal telling the computer "this is a color code" [cite: 35]. A standard code has 6 characters following the hash [cite: 36].

The RGB Mix [cite: 51]

- **RED (First 2 digits):** Controls the amount of red light [cite: 42, 43].
- **GREEN (Middle 2 digits):** Sets how much green light blends in [cite: 45, 46].
- **BLUE (Last 2 digits):** Decides the blue light intensity [cite: 47, 48].

Values range from **00** (none/silent) to **FF** (maximum brightness/loudest) [cite: 59].

4. Examples & Primary Colors

By mixing Red, Green, and Blue light in different amounts, we can create every color on a screen [cite: 49, 50].

- **Pure Red: #FF0000**  [cite: 65, 78]
- **Pure Green: #00FF00**  [cite: 69]
- **Pure Blue: #0000FF**  [cite: 73]
- **Coral Orange: #FF5733**  (Full Red, Med Green, Low Blue) [cite: 53, 57]

5. Activity: Build a Digital Rainbow

Follow these steps to create a spectrum of colors [cite: 86]:

1. **Red:** #FF0000 [cite: 87]
2. **Orange:** #FF7F00 [cite: 88]
3. **Yellow:** #FFFF00 [cite: 89]
4. **Green:** #00FF00 [cite: 90]
5. **Blue:** #0000FF [cite: 91]
6. **Violet:** #8B00FF [cite: 92]

6. Real-World Applications

Hex colors are used everywhere by digital creators [cite: 106]:

- **Website Designers:** Pick colors for buttons and backgrounds [cite: 107].
- **Game Developers:** Create colorful characters and worlds [cite: 108].
- **App Creators:** Make beautiful user interfaces [cite: 109].
- **Digital Artists:** Paint amazing artwork on computers [cite: 110].